

Eric Ponce

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Experience

E-Circuit Motors, Inc., R&D Director

Jan 2024 – Present | Bozeman, MT

- Lead the design and development of custom controller hardware (STM32) and firmware for BLDC motors that leverage core motor technology
- Drive innovation in advanced motor control strategies, including FOC, sensorless algorithms, parameter estimation, and torque optimization
- Architect software systems for interfacing with embedded controllers using a variety of communication protocols (CAN, UART, PWM, etc.)
- Implement trade show demos showcasing core motor technologies in real-world applications (robotics, haptics) to improve customer engagement
- Build custom infrastructure for high-power motor testing ($\approx 15\text{kW}$) by recycling power through a back-to-back setup
- Interface directly with keystone clients to diagnose and resolve complex technical issues, ensuring successful integration of company technology
- Prototype hardware and firmware to improve customer sales across all application areas (HVAC, pumps, UUV, robotics, manufacturing, consumer, etc.)
- Manage relationships with independent contractors and clients to accelerate product development and expand application reach
- Create custom motor, system, and control algorithm simulations with Python and QSPICE

MIT – Electromechanical Systems Group (Prof. Steven Leeb), Graduate Researcher

Jun 2017 – Sep 2023 | Cambridge, MA

- Developed physical models and low-power mixed-signal embedded circuitry (Infineon PSoC), firmware, and software for non-intrusively measuring the state and operation of mechanical systems, such as water meters, to cheaply retrofit for smart resource tracking and diagnostic capability
- Developed models, embedded circuitry (Infineon PSoC, Microchip SAML10), and firmware for harvesting ambient energy to enable sensing of systems
- Developed methods for analyzing the stability of grid-connected constant-power-loads which may cause serious instabilities
- Redesigned firmware and developed a new web software UI to drastically reduce power consumption and demonstrate new sensing capabilities for research group's wireless sensing platform centered around a TI Tiva C Series microcontroller and CC3100 Wi-Fi network processor
- Mentored/managed several undergraduate- and graduate-level researchers on development for research group topics and electrical engineering courses for students across a range of levels (middle-school through university)
- Teaching assistant for four semesters of Power Electronics and Microcomputer Lab. Managed staff hours, laboratory resources, and safety, maintained lab equipment, and helped students learn the material and debug their work. Achieved a TA rating of 6.96/7.0

Conduit Tech, Electrical Engineering Consultant

Jun 2022 – Aug 2022 | Cambridge, MA

- Developed hardware/firmware and defined sensor requirements for real-time AWS IoT Core-connected (Wi-Fi/NB-IoT) sensing platform for HVAC

WaterBit, Electrical Engineering Intern

Jun 2020 – Aug 2020 | San Jose, CA

- Developed real-time embedded signal processing methods to non-intrusively detect disastrous valve failure in harsh agricultural environments
- Restructured and rewrote ultra-low power firmware to enable development of new features and reduce technical debt
- Customized the firmware build system for our Infineon PSoC target to improve version control and continuous integration efforts

ORI, Electrical Engineering Consultant

Sep 2016 – Feb 2017 | Cambridge, MA

- Developed highly configurable and easily expandable embedded circuitry for control of movable furniture centered around an Infineon PSoC

Apple – iPad Systems, Electrical Engineering Intern

Jun 2016 – Aug 2016 | Cupertino, CA

- Developed software and tooling and interfaced with vendor to validate, debug, and improve early-stage USB power-delivery
- Validated power distribution and high-speed communication networks and interfaces
- Owned design and production of mixed-signal high-speed multi-layer PCB to enabled the testing of high pixel density LED displays
- Provided engineering support in Shenzhen during bring up of large-scale design verification build

Education – Massachusetts Institute of Technology

PhD in Electrical Engineering and Computer Science

GPA: 5.0/5.0

Jun 2019 – Sep 2023

MEng in Electrical Engineering and Computer Science

GPA: 4.8/5.0

Jun 2017 – Jun 2019

BS in Electrical Science and Engineering

GPA: 4.9/5.0

Aug 2013 – Jun 2017

Skills

Hardware	Analog, digital, embedded, low-power, and power electronic circuit design, KiCad, Altium, Cadence, SPICE, SolidWorks
Firmware	Communications (SPI, I2C, CAN, Wi-Fi, BT, NB-IoT, etc.), signal processing, multiple MCU architectures (8051, AVR, ARM Cortex-M), RTOS
Software	C, Python (NumPy, SciPy, Matplotlib), shell scripting (bash), Git, build system (GNU), HTML, CSS, JavaScript
Other	Numerical & analytical modeling, optimization, 3D printing, laser cutting, solder rework, Spanish

Publications

- **Ponce, E.**, & Leeb, S., “Impedance Modeling for Line-Frequency Rectified Constant-Power Loads,” *IEEE Access*, 2025
- **Ponce, E.**, & Leeb, S., “Envelope Impedance Modeling of Line Frequency Rectifiers,” *IEEE Trans. on Inds. Elec.*, 2025
- Monagle, D., **Ponce, E.**, & Leeb, S., “Rule the Joule: An Energy Management Design Guide for Self-Powered Sensors,” *IEEE Sensors*, 2023
- Monagle, D., **Ponce, E.**, & Leeb, S., “Resonant Circuits for Split-Core Magnetic Energy Harvesters,” *IEEE TIE*, 2023
- Monagle, D., **Ponce, E.**, & Leeb, S., “Generalized Analysis Method for Magnetic Energy Harvesters,” *IEEE TPEL*, 2022
- Gutierrez, M., Saathoff, E., **Ponce, E.**, & Leeb, S., “Sub Line-Freq. Stability Analysis of Single-Phase CPLs Using Envelope Impedance,” *IEEE TPEL*, 2022
- **Ponce, E.** & Leeb, S., “Passive Probe: Mechanically-Modulated Field Sensing for Motion Tracking and Flow Estimation,” *IEEE Sensors Journal*, 2022
- **Ponce, E.**, Leeb, S., & Lindahl, P., “Know the Flow: Non-Contact Magnetic Flow Rate Sensing for Water Meters,” *IEEE Sensors Journal*, 2021
- **Ponce, E.** & Leeb, S., “A 3-D Field Solution for Axially Polarized Multi-Pole Ring PMs and its Appl. in Position Meas,” *IEEE Trans. Magn.*, 2020